

Epidemic Data Visualization Surveillance Based on Flask

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Abstract: With the spread of the epidemic, the development of digital industry will be rapidly enhanced in the new situation and opportunities. The combination of mathematical model and graph makes it possible to predict the trend of infectious diseases according to the different transmission speed, spatial scope, transmission route diversity, dynamic mechanism and other factors. The visualization technology of infectious disease model data also plays an important role in epidemic data detection. In this paper, real-time monitoring, quantitative analysis, dynamic prediction and assessment of the current severe situation of the COVID-19 epidemic were conducted to obtain relevant information on the development of the epidemic, objectively estimate the current situation of the COVID-19 epidemic, and predict the development trend in the future. The innovative drag-and-drop recalculation, data viewing, distance roaming and other functions in this paper have greatly improved user experience and enabled users to have the ability of data mining and integration. It provides a one-stop solution, which is from the template, ORM, Session and the Authentication background. It is very convenient to use. The monitoring and trend prediction platform has also naturally become a powerful helper for the government to realize comprehensive monitoring and decision-making on COVID-19.

Keywords: Epidemic monitoring, Data Visualization, Flask Framework, Web Platform

I. Introduction

With the improvement of digital level, data visualization technology can be used to display the changes of data intuitively. The establishment of mathematical models can provide important decision-making basis for prediction and early warning. In this epidemic situation, the establishment of epidemic visualization platform and epidemic trend prediction research are of great significance to the analysis of disease transmission ability and the prediction of transmission development trend. It can provide valuable data support for the epidemic prevention and control center to formulate corresponding intervention and prevention and control measures [1].

Starting from 1 January 2020, the European Centre for Disease Prevention and Control (ECDC) will automatically collect data on cases diagnosed with coronavirus -19 and mortality. It has developed Shiny applications for data visualization and multi-indicator analysis to track the sarS-COV-2 epidemic using ECDC data [2]. It provides basic country-specific epidemiological surveillance tools in an interactive and user-friendly manner. Available analyses include temporal trends and projections, morbidity, population mortality, mortality and basic reproductive figures [3]. The

GLOBAL Epidemic Prediction System will continue to be updated and optimized. In the future social development, the system will no longer be limited to the prediction of COVID-19, but will develop towards the prediction of global epidemic epidemiology [4].

Since the outbreak of the epidemic, a large number of epidemic data has appeared on major online platforms. Obtaining accurate epidemic data and timely display is a major feature of the project. In addition, in the improvement of the epidemic trend prediction model, there are two groups of new deaths and asymptomatic infected people. The optimization and improvement of the traditional model is another feature [5]. The parameters of the prediction model are set based on experts' opinions and reports, and the prediction trend is closest to the actual trend. Therefore, the characteristics and transmission routes of epidemic diseases and the intervention measures of prevention and control departments should be taken into consideration when selecting the prediction model of infectious diseases. It promotes the combination of COVID-19 development and information technology, and then provides valuable data support for the formulation of corresponding intervention and prevention and control measures [6].

In this paper, the framework of Flask is used. The Python crawler technology is adopted to crawl the historical epidemic data and the current epidemic data on the website. Mysql is used to store the obtained data. After data preprocessing and cleaning, the association and shortcomings of the model are found out, and the improved model scheme is proposed. The training data is used to optimize the model to solve the problem that ordinary parsing technology can not resolve. After the optimal model is obtained, the epidemic trend is predicted and analyzed, and the theoretical research and analysis are obtained.

II. Epidemic data visualization model

The traditional SEIR model mainly uses the differential variance to show the temporal changes of the four groups of people under relatively loose conditions. However, in practical applications, the more detailed the classification of parameters and features is, the more accurate the trend prediction of the model is [7]. Uncertainties can lead to large fluctuations in the accuracy of predictions, such as the emergence of asymptomatic patients in various regions in the middle of the epidemic, which has a great impact on the development of the epidemic trend. Based on the above situation, on the basis of the original SEIR model, the new parameters of A

asymptomatic infected person and D dead person are added. The modeling flow chart is shown in Figure 1.

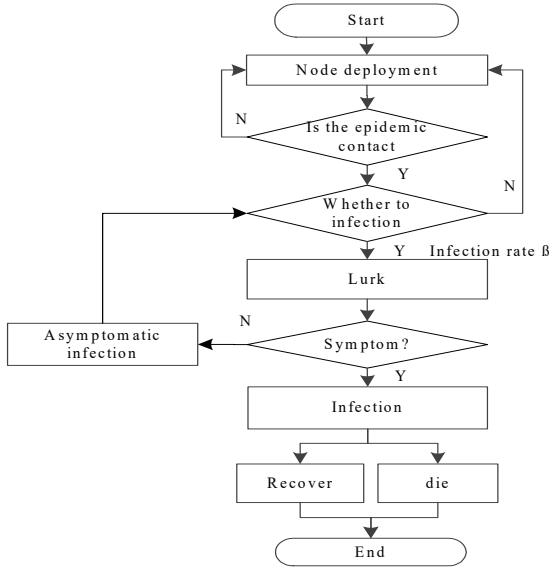


Figure 1 Modeling flow chart of SEIARD model

SEIADR (Susceptible Exposed Infectious Asymptomatic Death Recovered)

SEIADR (Susceptible Exposed Infectious Asymptomatic Death Recovered) model divides the population into six categories:

(1) Susceptible population (S): The body immunity is very low. In the exposed environment, it is very easy to be infected after contact with the infected person. The process of $S \rightarrow E$ is determined by the contact rate α and the infection rate β [8].

(2) Latent population (E): Those who have been in contact or close contact with the infected person, also known as close contacts, have not yet shown the clinical manifestations of the infected person within a certain incubation period, but have the virus in the body and have the ability to infect.

(3) Infected people (I): Evolved from the latent, the body has been infected by the virus, has emerged clinical symptoms and has a strong ability to infect [9].

(4) People with asymptomatic infection (A): In the incubation period, there has been no manifestation of clinical symptoms, and after the incubation period, there has been no manifestation of clinical symptoms. But it has been monitored to be positive. There is already a virus in the body, and it has the same strong infectious ability as the infected person.

(5) Dead people (D): It refers to people who died because they were infected with the virus and were not cured.

(6) Convalescent people (R): It refers to people who are negative after treatment, have been cured or self-healing, and have immunity in the future for a period of time, and will not appear to be positive.

Figure 2 shows the virus transmission process in the SEIADR model.

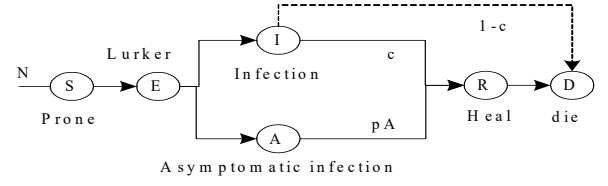


Figure 2 SEIADR virus transmission process diagram

In the process diagram shown in Fig. 2, D represents the number of deceased persons. A represents the number of asymptomatic infected persons. α represents the contact rate. $1-c$ represents the probability of death of infected persons, and ρ represents the recovery rate of asymptomatic infected persons. γ is the latency probability, and $1-\gamma$ is the conversion rate of the asymptomatic population [10]. According to the propagation process, the differential equation of the model can be expressed as:

$$\frac{dS}{dt} = -\alpha\beta\frac{SI}{N} - \alpha\beta\frac{SA}{N} \quad (1)$$

$$\frac{dE}{dt} = \alpha\beta\frac{SI}{N} + \alpha\beta\frac{SA}{N} - \gamma E - (1-\gamma)E \quad (2)$$

$$\frac{dI}{dt} = \gamma E - cI \quad (3)$$

$$\frac{dA}{dt} = (1-\gamma)E - \rho A \quad (4)$$

$$\frac{dD}{dt} = (1-c)I \quad (5)$$

$$\frac{dR}{dt} = cI + \rho A - (1-c)I \quad (6)$$

In the classical model, the effect of vaccine is included, and the prediction of the model will also have different effects. The virus has an incubation period, that is, Exposed, represented by E , which is the classic SEIR model. The patient in the incubation period is also infectious, but the infectious ability is lower than that of the infected person. The susceptible population begins to vaccinate, and the immune effect of the vaccine is between 0 and 1. Therefore, the differential equation of the model will increase the speed of infection from the infected person to the susceptible person (not vaccinated), the speed of infection from the latent person to the susceptible person (not vaccinated), the speed of transmission from the infected person to a susceptible person (vaccinated but not immunized), and the rate of transmission from a latent person to a susceptible (vaccinated but not immune).

Based on the traditional model, the differential equation after considering the vaccine can be obtained:

$$\frac{dS}{dt} = -\frac{(1-\alpha)\beta IS}{N} - \frac{(1-\alpha)\beta SqE}{N} - \frac{(1-p)I\beta aS}{N} - \frac{(1-p)\beta aSqE}{N} - p\alpha S \quad (7)$$

$$\frac{dE}{dt} = \frac{(1-\alpha)\beta IS}{N} + \frac{(1-\alpha)\beta SqE}{N} + \frac{(1-p)I\beta aS}{N} + \frac{(1-p)\beta aSqE}{N} - eE \quad (8)$$

$$\frac{dI}{dt} = eE - \gamma I \quad (9)$$

$$\frac{dR}{dt} = p\alpha S + \gamma I \quad (10)$$

Considering that there are many influencing factors of infectious diseases, and the effective regeneration number varies with the government's epidemic prevention measures, isolation and vaccination, the parameter setting of the prediction model after considering the addition of vaccines needs to be further explored.

III. Experimental analysis

A. Experimental data

This paper uses the MySQL database and the interaction between Python and MySQL, which stores the persistent data in the database for the subsequent platform display. Then the Flask framework of Python is used to quickly build a Web platform. The HTML5, CSS and other front-end technologies and various curve modules of Echarts library are used to draw visual panels to achieve the display of epidemic data monitoring. Flask is a lightweight Web application framework (Python) that is easy to encapsulate and not complex in logic. The WSGI (Python Web Server Gateway Interface) toolkit is implemented using Werkzeug. The framework's template

engine uses jinja2 [11]. Below through the interface design, routing design, application modular design, configuration and other four aspects of the two frameworks will be compared to select the best framework for this paper.

B. Improved model evaluation and analysis

According to the Novel Coronavirus epidemic data of the National Health Commission obtained through the web crawler, the training data of the prediction model from January 20, 2021 to April 1, 2021 is taken as the training data. After deaths and asymptomatic infected persons are introduced, the trend curve of the model is generated according to the differential equation of the improved model and compared with the basic model. Then, analyze the characteristics of the improved model in the trend prediction [12].

(1) Combine the basic model of SEIR model and the improved model SEIARD, and use the data to fit the curve of the actual confirmed number trend and the model prediction trend, as shown in Figure 3.

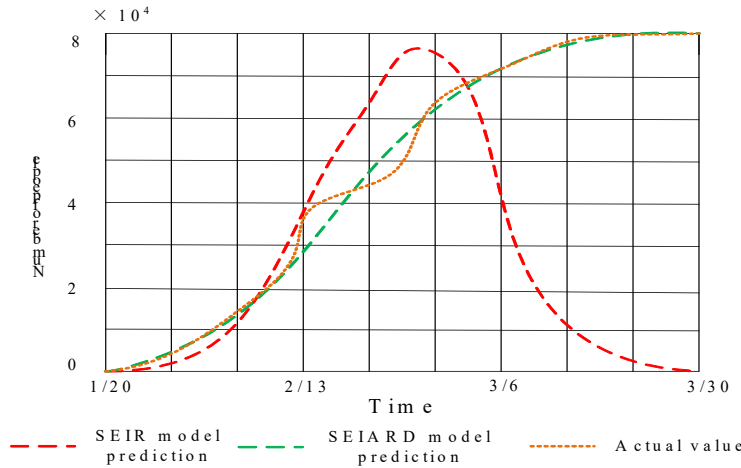


Fig. 3 Model fitting results

SEIR model fitting results have relatively accurate prediction effect in the early stage of the epidemic, but there is a large number of reports with the actual number of confirmed cases. The simulated prediction trend curve shows a trend of first increase and then decrease, and the effect is not ideal. The prediction curve fitted by the improved SEIARD model can be seen in the report. The cumulative number of confirmed cases will gradually reach the peak at the end of February and early

March, and tend to stabilize in the middle of March. The COVID-19 epidemic will enter a stable period. The result basically coincides with the data curve of the actual number of confirmed cases. By the end of March, if the government takes appropriate epidemic prevention and control measures, China will soon enter a new stage of resuming work and production, and people will gradually be able to return to normal life.

(2) Combine the basic model of the SEIR model and the improved model SEIARD, and use the data to fit the curve of

the actual death toll trend and the model prediction trend, as shown in Figure 4.

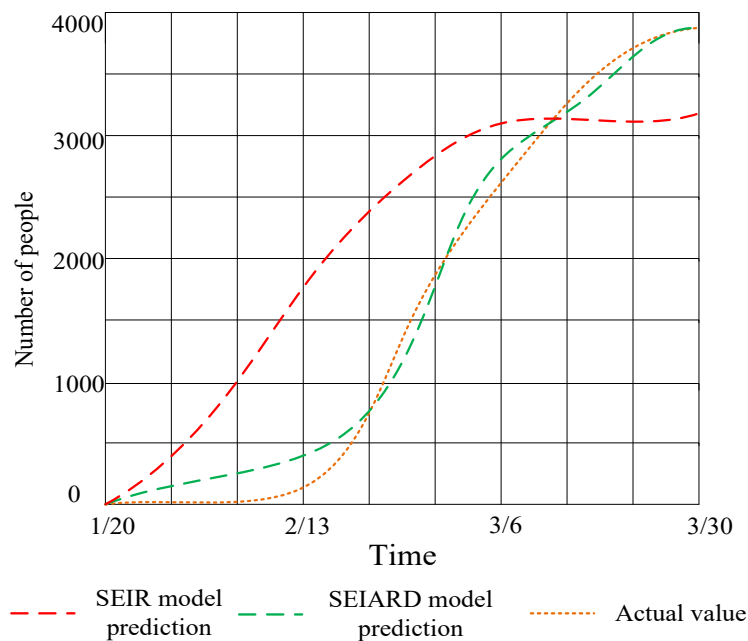


Fig. 4 Fitting results of SEIARD model

The fitting results of SEIR model have a great impact on the actual number of confirmed cases. The peak time of the simulated prediction trend curve is later than the actual peak time, which has exceeded the reasonable error range, and the effect is not ideal. The prediction curve fitted by the improved model SEIARD can be seen that the curve is only slightly deviated in the middle of February. But within a reasonable error range, the other time basically coincides with the actual curve. The time peak of the actual death number curve is basically consistent with the time peak of the improved prediction model, that is, in the middle and early March. It can provide favorable theoretical value for epidemic prevention and control departments, and facilitate departments to formulate more effective measures to deal with the current epidemic situation.

After comparing the curve trends of the cumulative number of deaths and the cumulative number of confirmed cases in the basic model and the improved model, the following conclusions can be drawn:

(1) Because the traditional SEIR model does not take into account the dead and asymptomatic patients, when drawing the trend curve, it can only predict the general development trend of the epidemic, but can not accurately estimate the peak time point of the cumulative confirmed or cumulative death toll. Besides, it can not provide accurate time points for the key trend nodes of the epidemic. Therefore, it is not in line with the actual development of the epidemic.

(2) The trend of the improved SEIARD model is basically consistent with the curve of the actual cumulative number of

confirmed cases and the actual cumulative number of deaths after the new deaths and asymptomatic infections are added. In the fitting results, we can see that in mid-January and mid-February, the cumulative confirmed cases showed an exponential increase, indicating that people began to pay attention to it. The ability of nucleic acid detection was also greatly improved. In addition, the number of people in the incubation period is gradually increasing, and a large number of asymptomatic infections will appear. Prevention and control departments can formulate more effective prevention and control measures based on these data and information. Therefore, within the allowable range of reasonable error, the improved SEIARD model can provide a highly accurate prediction scheme in predicting the key trend nodes of the epidemic situation, and is expected to become an effective helper of the epidemic prevention and control center and other prevention and control departments.

IV. Conclusion

In this paper, the visualization of time series data is selected to develop the visual surveillance of epidemic situation. The conclusions are as follows:

- (1) The crawling epidemic data on a website through a data crawler technology is adopted to serve as a training data set and a test data set.
- (2) An improved model scheme based on that association of the basic model is proposed.

(3) The data model is evaluated and optimized. Then, compare and verify the experimental data, and finally make theoretical analysis.

The design of this paper meets the initial design requirements of the project, but there is still room for improvement. In the analysis of the improved prediction model, the research results show that the improved prediction model is reliable in the trend prediction, which can provide some theoretical support for the prevention and control departments to formulate epidemic intervention measures in the future.

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